

Ishaan Goel

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Education

Illinois Institute of Technology (3.6/4.0)	Aug 2022 – May 2027
Master of Artificial Intelligence	
Bachelor of Science in Applied Mathematics	
Bachelor of Science in Data Science	

Experience

Institute for Pure and Applied Mathematics, UCLA	Los Angeles, CA
<i>Undergraduate Researcher, RIPS REU, sponsored by the Aerospace Corporation</i>	Jun 2026 – Aug 2026
Developed an Extended Kalman Filter fusing ADS-B, computer vision, and radar measurements for aircraft surface localization, reducing RMSE by 59.7% over ADS-B alone and matching the accuracy of ASDE-X, a \$550M+ FAA surveillance system, at orders of magnitude lower cost.	
Illinois Institute of Technology, Department of Applied Mathematics	Chicago, IL
<i>Research Assistant under Dr. Igor Cialenco, NSF-funded research</i>	Aug 2024 – Present
Developing a multi-agent reinforcement learning framework for stochastic groundwater markets under California's SGMA to analyze generalized Nash equilibrium behavior, constraint satisfaction, and policy outcomes under coupled physical and regulatory constraints.	
PeopleStrong Technologies Ltd.	Gurugram, India
<i>Artificial Intelligence Engineering Intern</i>	Jun 2025 – Jul 2025
Built a scalable RAG-based system for Automated Ticket Resolution, enabling document-grounded Q&A for 1,000+ employees and reducing index testing time by over 50x (weeks to hours). Deployed production-ready agentic AI systems and accelerated iteration cycles through automation and containerization.	
<i>Artificial Intelligence Engineering Intern</i>	Jul 2024 – Jul 2024
Designed and deployed a scalable LLM-based chatbot with an agentic architecture integrating 5 RAG systems, validated via A/B testing and approved for company-wide rollout.	

Relevant Coursework

Math: Probability, Linear Optimization, Stochastic Processes, Graph Theory, Monte Carlo Methods in Finance, SPDEs, Real Analysis, Abstract Algebra, Combinatorics, Dynamical Systems, Functional Analysis (audit)
AI/ML: Machine Learning, Statistical Learning, Natural Language Processing, Artificial Intelligence, Data Mining, Deep Learning, Information Retrieval, Algorithms, Database Organization, Big Data, Game Theory (audit)
In Progress: Social Graph Analysis & Graph Learning, Agentic AI for Campus Innovation

Projects

Full-Stack Information Retrieval Engine <i>IR, scikit-learn, Flask, Scrapy, Python</i>	Oct 2025 – Dec 2025
Developed a full-stack information retrieval engine featuring a Scrapy crawler with configurable depth and AutoThrottle, a TF-IDF inverted index with cosine similarity and optional semantic k NN search (word2vec, FAISS), and a Flask-based query processor with Top-K ranking, NLTK spell correction, WordNet expansion, and CSV export.	
Distributed RL for Plastic Drift Simulation <i>RL, PyTorch, JAX, Python</i>	Aug 2025 – Nov 2025
Developed a distributed Monte Carlo RL framework that simulates ocean plastic drift and aggregates results from volunteer clients for large-scale model optimization, earning 3rd place in the Grainger Computing Innovation Prize.	

Skills

Programming: Python, R, Julia, C++, Java, $\text{\LaTeX}$, HTML, CSS, SQL, Mathematica, MATLAB
Python Stack: PyTorch, scikit-learn, NumPy, pandas, matplotlib, NetworkX, PySpark
Spoken Languages: English, Hindi, Urdu, Mandarin Chinese, Spanish, Punjabi, Modern Standard Arabic
Others: Generative AI, Operations Research, Stochastic Modeling, Discrete Mathematics, Mathematical Finance

Leadership Roles

Vice President , SIAM Chapter at Illinois Institute of Technology	Dec 2022 – Present
Organized networking events for students across multiple disciplines related to mathematics. Connected students to research opportunities in applied mathematics. Hosted professor- and student-led talks.	
President , Machine Learning at Illinois Institute of Technology (ML@IIT)	Jan 2023 – May 2025
Organized workshops and speaker events on AI/ML. Scaled organization from small group to 300+ active participants.	